Skills Summary

Parametric, Polar, and Vector-Valued Calculus

Use this reference while completing your worksheet or when studying.

Every question here is one of four: a *slope*, a *magnitude*, an *intersection*, or an *accumulation*. Name which one before you write anything — that choice, not the algebra, is what these problems test.

1. Parametric slope (parametric-derivative)

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \qquad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$$

Horizontal tangent: $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} \neq 0$. Vertical tangent: $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} \neq 0$. Both zero: inconclusive, examine separately.

Example: $x = t^2 + t$, $y = t^3 + 2t$. Find $\frac{dy}{dx}$ at $t = 1$.
Step 1. The curve is given through a parameter, so the slope is the ratio of the two t-derivatives — differentiate each separately first.
Step 2. $\dfrac{dy}{dx} = \dfrac{3t^2+2}{2t+1}$, and at $t = 1$ this is $\dfrac{5}{3}$. $\boldsymbol{\left.\frac{dy}{dx}\right|_{t=1} = \frac{5}{3}}$

Try it: $x = 2t^2$, $y = t^3 - t$. Find $\frac{dy}{dx}$ at $t = 2$.

check: $\boldsymbol{\frac{11}{8}}$

2. Vector motion: velocity, speed, acceleration

$\mathbf{v}(t) = \langle x'(t),\, y'(t)\rangle$, speed $= |\mathbf{v}(t)| = \sqrt{x'(t)^2 + y'(t)^2}$, $\mathbf{a}(t) = \langle x''(t),\, y''(t)\rangle$.
The particle is **speeding up** when $\mathbf{v} \cdot \mathbf{a} > 0$ and slowing down when $\mathbf{v} \cdot \mathbf{a} < 0$ — direction, not the sign of $\mathbf{a}$ alone.

Example: $\mathbf{r}(t) = \langle t^2 - t,\ t^3\rangle$. Find the speed at $t = 1$.
Step 1. Speed is a magnitude, so build the velocity vector first and then take its length.
Step 2. $\mathbf{v}(t) = \langle 2t-1,\ 3t^2\rangle$, so $\mathbf{v}(1) = \langle 1, 3\rangle$ and $|\mathbf{v}(1)| = \sqrt{1+9}$. $\boldsymbol{\sqrt{10} \approx 3.162}$

Try it: $\mathbf{r}(t) = \langle 3t^2,\ t^3 - t\rangle$. Find the speed at $t = 2$.

check: $\boldsymbol{\sqrt{265} \approx 16.279}$

3. Where two polar curves meet (polar-intersection)

Set $r_1(\theta) = r_2(\theta)$ and solve on $[0, 2\pi)$ — that finds the points reached at the *same* angle. Then check the pole separately: if each curve hits $r = 0$ at *some* angle, the curves also meet at the origin even though the angles differ.

Example: Where do $r = 1 + \cos\theta$ and $r = 3\cos\theta$ meet?
Step 1. Both radii are functions of θ, so equal radius at equal angle is the condition to solve.
Step 2. $1 + \cos\theta = 3\cos\theta \Rightarrow \cos\theta = \frac{1}{2}$. $\boldsymbol{\theta = \frac{\pi}{3}, \frac{5\pi}{3}}$

Try it: Find every θ in $[0, 2\pi)$ where $r = 2\sin\theta$ meets $r = 1$.

check: $\boldsymbol{\theta = \frac{\pi}{6}, \frac{5\pi}{6}}$

4. Accumulated quantities (accumulated-quantity)

$$\text{distance} = \int_a^b |\mathbf{v}(t)|\,dt, \qquad \text{polar area} = \frac{1}{2}\int_\alpha^\beta r^2\,d\theta, \qquad \frac{1}{2}\int_\alpha^\beta \left(r_{\text{out}}^2 - r_{\text{in}}^2\right)d\theta$$

Arc length of a parametric curve is the same integral as distance travelled. Choose limits that trace the region or path exactly *once*.

Example: Area of one petal of $r = 2\sin(3\theta)$.
Step 1. This is an accumulation of area, so use $\frac{1}{2}\int r^2\,d\theta$ over one petal: θ from 0 to $\frac{\pi}{3}$, where r returns to 0.
Step 2. $\frac{1}{2}\int_0^{\pi/3} 4\sin^2(3\theta)\,d\theta = \int_0^{\pi/3}(1 - \cos(6\theta))\,d\theta = \frac{\pi}{3}$. $\boldsymbol{\frac{\pi}{3} \approx 1.047}$

Try it: A particle has $\mathbf{v}(t) = \langle 2t,\ 2t^2 \rangle$. Find the distance travelled on $0 \le t \le 3$.

check: $\boldsymbol{\frac{2}{3}(10\sqrt{10} - 1) \approx 20.415}$